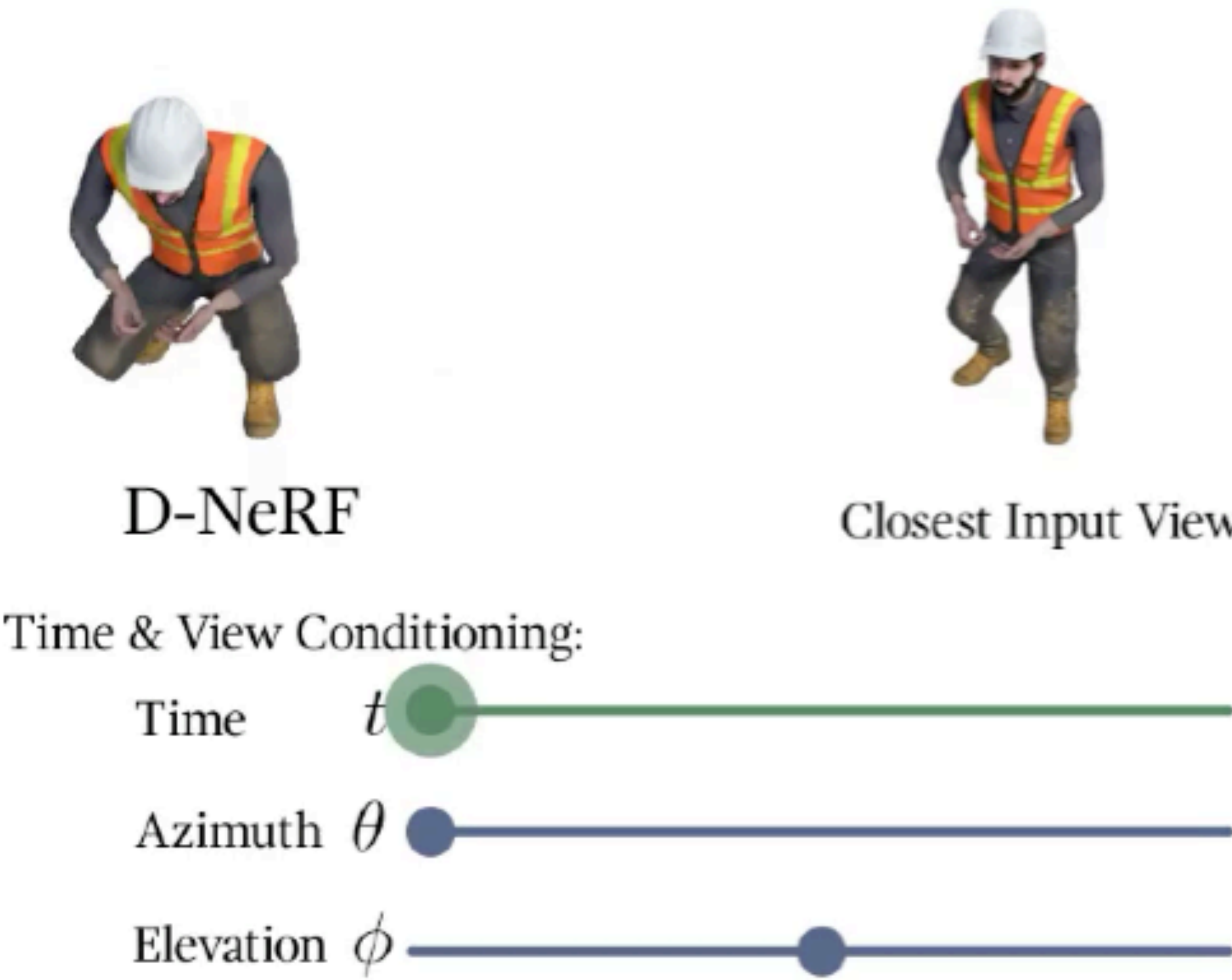
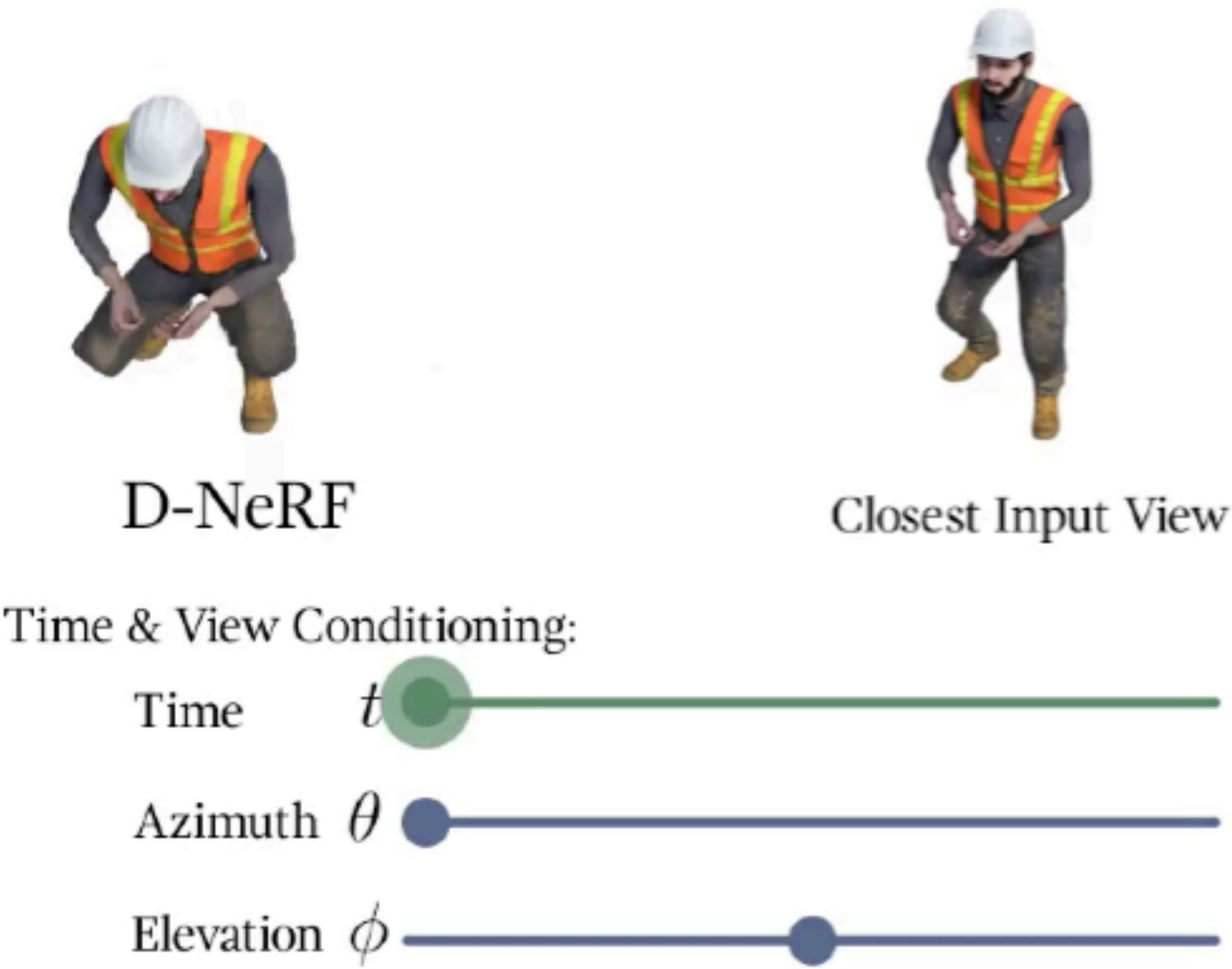
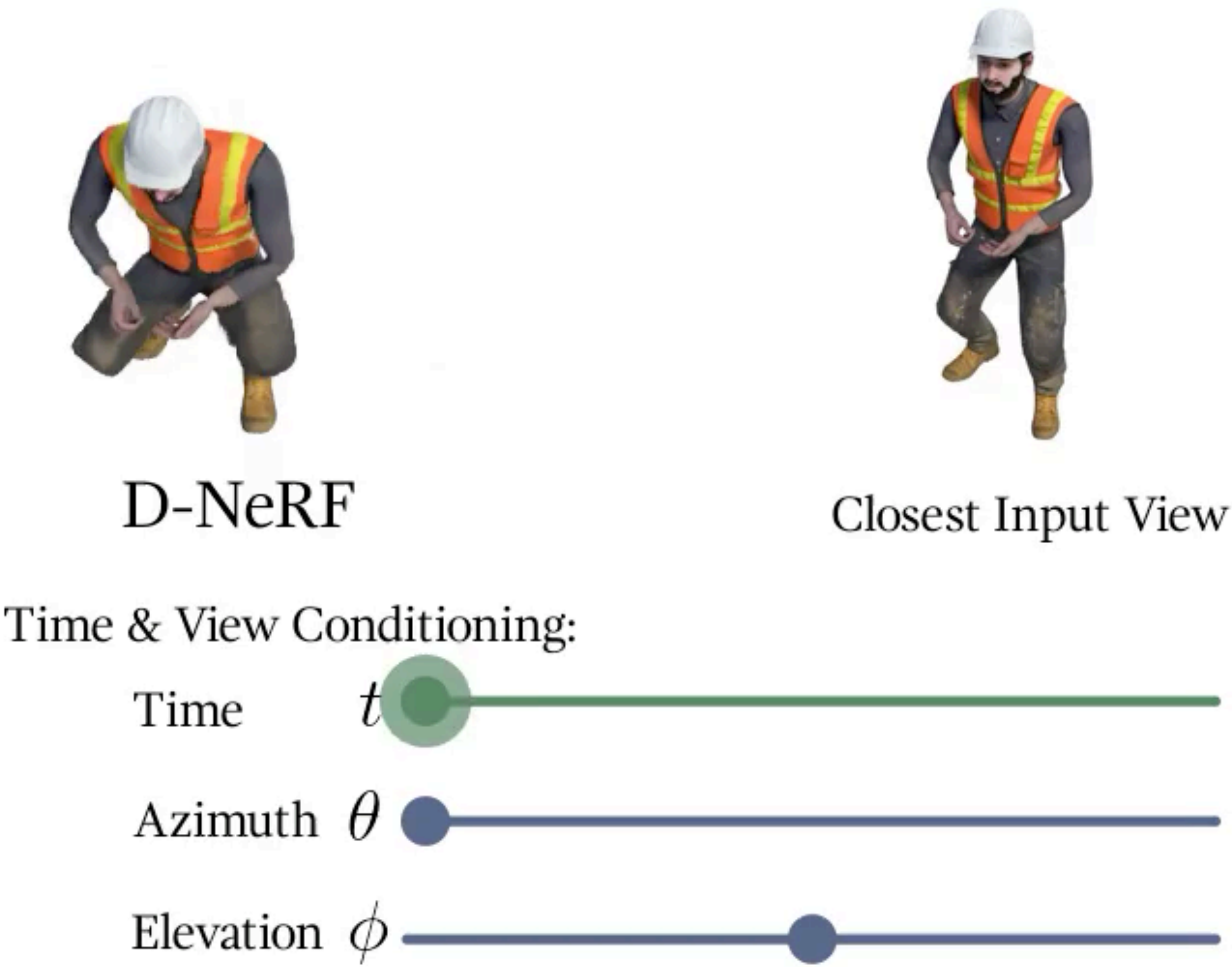
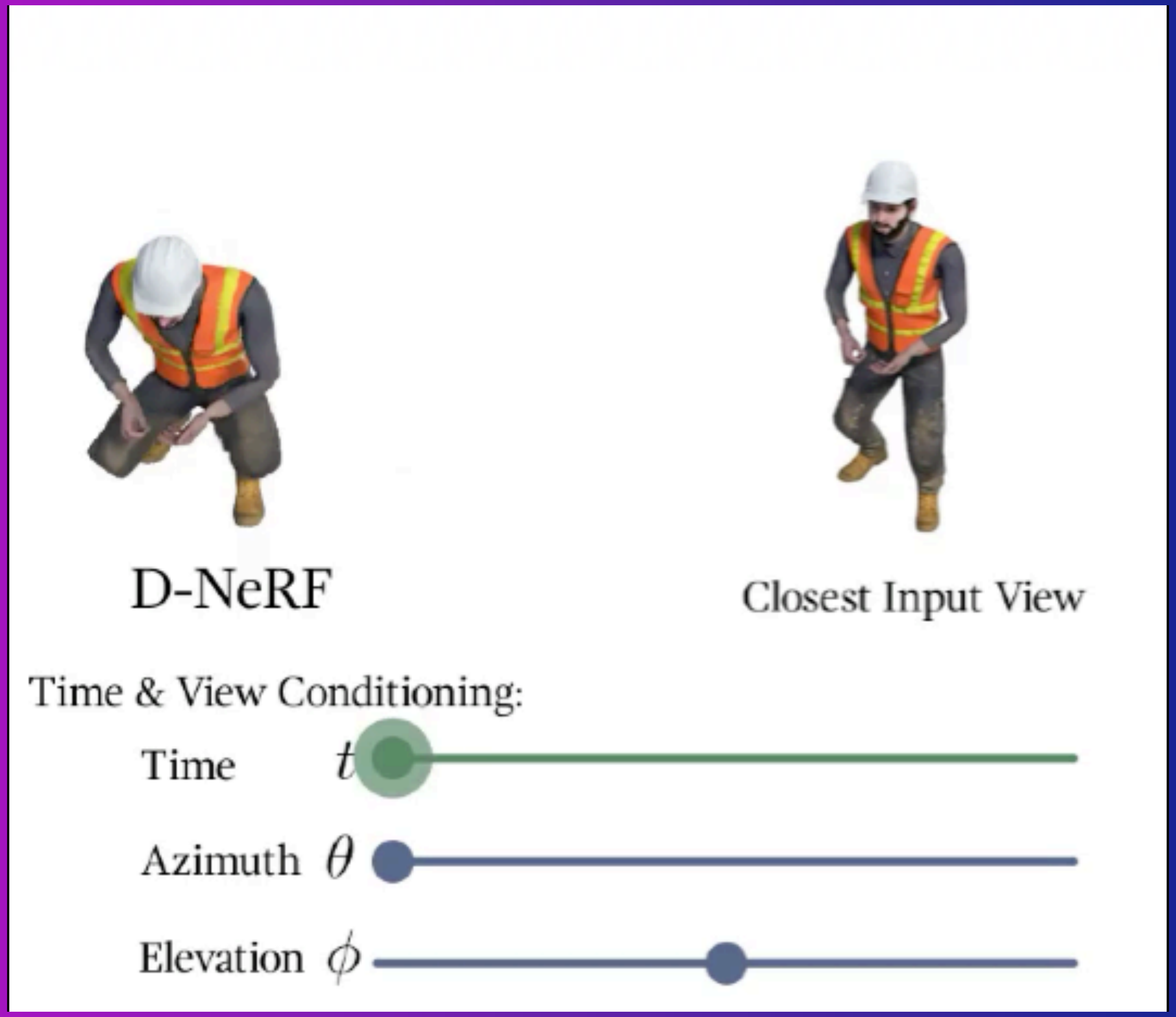

D-NeRF
Closest Input View
Time & View Conditioning:
Time t
Azimuth θ
Elevation ϕ


D-NeRF
Closest Input View
Time & View Conditioning:
Time t
Azimuth θ
Elevation ϕ


D-NeRF
Closest Input View
Time & View Conditioning:
Time t
Azimuth θ
Elevation ϕ

Physics-based Learning: NeRF

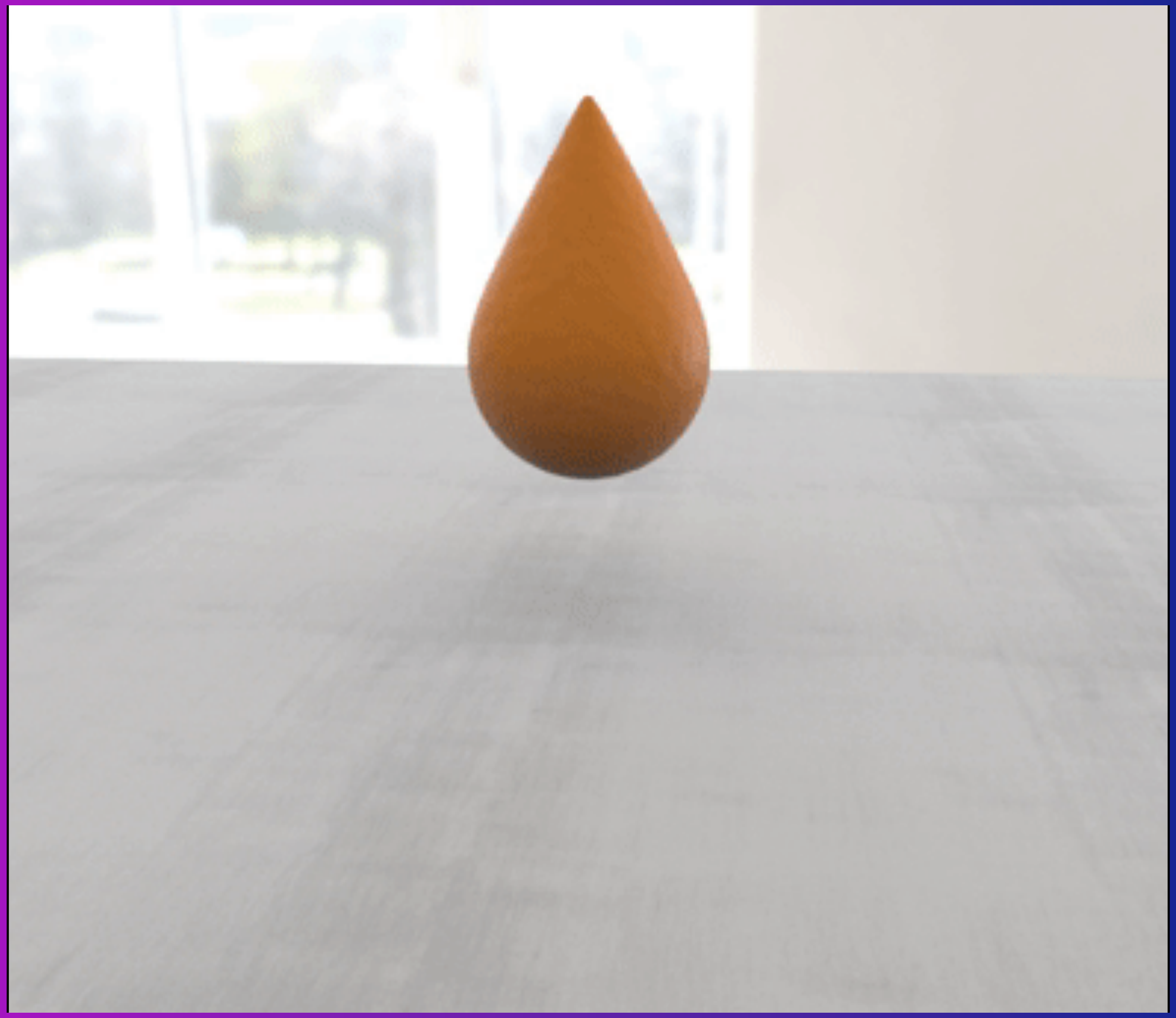
D-NeRF



Dynamic Neural Radiance Field

- Pumarola et. al. 2021.
- Novel spatial and temporal view synthesis

PAC-NeRF



Physics-Augmented Continuum

- Li et al. 2023.
- Newtonian physics simulations.
- Advect via MPM.

David Levin
University of Toronto
Dept. Computer Science
NVIDIA Toronto

- Physics-augmented novel view synthesis for dynamic scenes

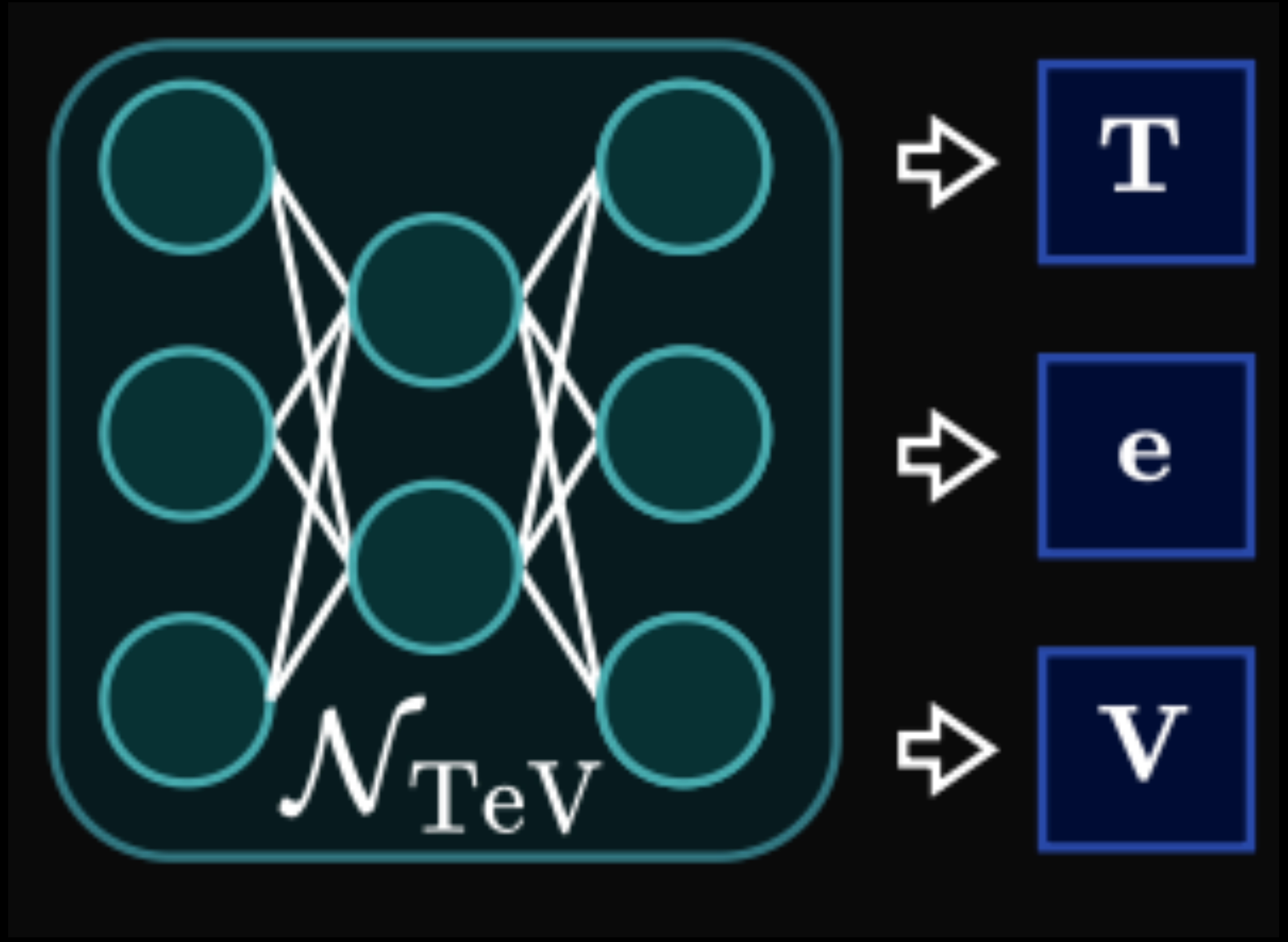
$$\frac{D\sigma}{Dt} = \frac{Dc}{Dt} = 0$$

- D is material derivative

$$\frac{D\phi}{Dt} = \frac{\partial \phi}{\partial t} + \mathbf{v} \cdot \nabla \phi$$

Physics-based Learning: Infrared Generation

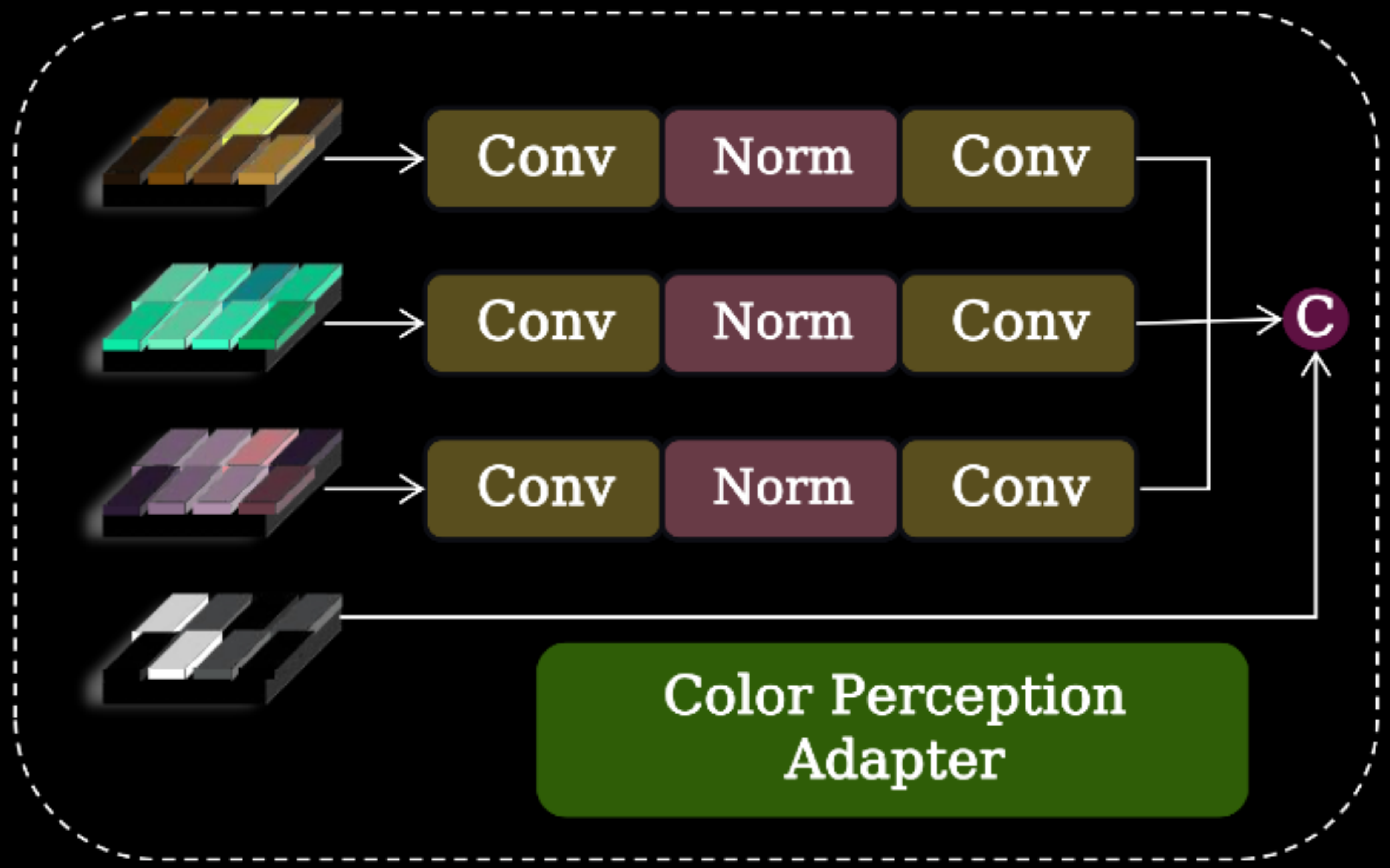
Diffusion Implementation



Physics-informed Diffusion for IR Generation

- Fangyuan Mao et. al. (July 2024).
- Introduce TeV Decomposition Network.
- Distinct architectural component - deploy in other settings.

Transformer Implementation



Physics-based Preprocessing

- Improve performance via image preprocessing.
- Deploy Enhanced Perception Attention Module.
- Transformer deals with global rendering.